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Surveillance of respiratory viruses in severe acute respiratory infections in Southern Brazil, 2023–2024

Amanda Pellenz Ruivo[1], Milena da Cruz Bauermann[2], Tatiana Schäffer Gregianini[3], Franciellen Machado dos Santos[1], Fernanda Godinho[3], Ludmila Fiorenzano Baethgen[3], Taís Raquel Marcon Machado[3], Leticia Garay Martins[3], Renata Petzhold Mondini[3], Carolina Nunes Port[3], Artur Correa[1], Taina Selayaran[1], Paola Cristina Resende[4], Gabriel da Luz Wallau[5,6,7], Richard Steiner Salvato[1,3,8*†] and Ana Beatriz Gorini da Veiga[1†]

Abstract

Severe Acute Respiratory Infection (SARI) is one of the leading causes of death worldwide, representing a significant public health challenge. Respiratory viruses are the primary pathogens responsible for these infections, and their ability to evolve and spread efficiently contributes to their widespread circulation. Here, we examined the epidemiological characteristics of SARI cases reported over one year (February 2023 to February 2024) in Rio Grande do Sul, the southernmost state in Brazil. Additionally, 4,000 negative specimens for influenza, Respiratory Syncytial Virus (RSV), and SARS-CoV-2, tested routinely, were evaluated by an expanded PCR respiratory panel (11 pathogens). During the study period, 14,816 SARI cases were reported: 3,396 cases due to SARS-CoV-2 infection, 2,329 to RSV, 1,124 due to influenza (802 influenza A and 322 influenza B), and 7,803 due to undefined etiological agents. Differential diagnostics allowed for identifying at least one respiratory pathogen in 1,741 (43.5%) of the 4,000 SARI cases tested. Human rhinovirus was the most frequently detected virus (in 63.3% of positive cases), followed by metapneumovirus (17.7%), parainfluenza virus (15.9%), adenovirus (13.6%), and bocavirus (11.6%). Our findings highlight the significant role of a diverse range of respiratory viruses in contributing to severe illness and mortality. Additionally, we observed ongoing shifts in the seasonal patterns of these viruses compared to years before the COVID-19 pandemic, underscoring the dynamic nature of respiratory virus circulation. These results emphasize the need for comprehensive interventions to mitigate the impact of respiratory infections and ensure an adequate public health response.

Keywords Severe acute respiratory infection, Viruses, Public health, Brazil

†Richard Steiner Salvato and Ana Beatriz Gorini da Veiga contributed equally to this work.

*Correspondence:
Richard Steiner Salvato
richardssalvato@gmail.com
[1]Programa de Pós-graduação em Biociências, Universidade Federal de Ciências da Saúde de Porto Alegre, Porto Alegre, Brazil
[2]Universidade Luterana do Brasil, Canoas, Brazil
[3]Centro Estadual de Vigilância em Saúde, Secretaria Estadual de Saúde do Rio Grande do Sul (CEVS/SES-RS), Porto Alegre, Brazil
[4]Laboratório de Vírus Respiratórios, Enterovírus e Emergências Virais, Instituto Oswaldo Cruz, Fundação Oswaldo Cruz, Exantemáticos, Rio de Janeiro, Brazil
[5]Instituto Aggeu Magalhães, Fundação Oswaldo Cruz, Recife, Brazil
[6]Department of Arbovirology and Entomology, Bernhard Nocht Institute for Tropical Medicine, Hamburg, Germany
[7]Universidade Federal de Santa Maria, Santa Maria, Brazil
[8]Universidade Federal de Ciências da Saúde de Porto Alegre, Rua Sarmento Leite 245, Porto Alegre CEP: 90050-170, Rio Grande do Sul, Brazil

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Introduction

Acute respiratory infections are among the leading causes of mortality worldwide, accounting for over 4 million deaths estimated annually [1]. Severe Acute Respiratory Infection (SARI) is a serious respiratory condition characterized by an acute infection with symptoms within 10 days of infection, cough, fever, and hospitalization. SARI has long been recognized as a significant cause of morbidity and mortality worldwide, posing an important public health challenge [2].

Viruses, the primary pathogens responsible for respiratory infections, including SARI cases, are significant contributors to global morbidity and mortality, and recent evidence suggests that climate change may further increase the risk of viral emergence and transmission by influencing viral biology, host susceptibility, human behavior, and environmental conditions [3]. Respiratory viruses are characterized by high transmissibility and their ability to rapidly adapt, contributing to their impact on public health. The most commonly studied viruses associated with SARI include influenza virus A (IAV) and B (IBV), SARS-CoV-2, and respiratory syncytial virus (RSV), all of which circulate within the community and contribute to a significant number of hospitalizations, particularly in children and the elderly [4]. Despite the availability of vaccines and antivirals, these viruses continue to pose significant challenges to global public health [4].

Other viruses also target the respiratory epithelium and can contribute to the development of SARI, particularly in young children, immunocompromised individuals, and the elderly. These include human parainfluenza viruses (hPIV), human metapneumovirus (hMPV), human rhinovirus (hRV), human bocavirus (hBoV), human adenovirus (hAdV), respiratory enterovirus (EV), human parvovirus B19 (B19V), and other endemic human coronaviruses such as 229E, OC43, NL63, and HKU1. The wide diversity of these viruses and the challenges laboratories encounter in performing comprehensive testing lead to several undiagnosed cases [5, 6]. Despite recent advances in molecular diagnosis, including Next Generation Sequencing-based approaches that have become more accessible and increasingly implemented in public health laboratories, integrating broader respiratory virus diagnostics into routine surveillance remains a significant challenge in many resource-limited settings. Key barriers include the high cost of reagents and equipment, the need for trained personnel, and limitations in laboratory infrastructure [7]. As a result, national surveillance programs often prioritize testing for viruses with the greatest known epidemiological impact, which can hinder the detection of emerging or neglected respiratory pathogens [8].

In Brazil, over 264,000 SARI cases and 20,000 related deaths were reported in 2024, with 49% of the cases lacking an identified etiological agent [9]. Rio Grande do Sul, the southernmost state of Brazil, is characterized by cold winters; hence, people tend to stay indoors in less ventilated places, which contributes to increased transmission of respiratory viruses [10]. In 2024, 16,400 SARI cases were reported in the state, with 1,457 deaths, and approximately 50% of these cases remained without identification of an etiological agent, posing a significant challenge to implementing effective public health strategies for disease management and mitigation [11].

Here, we analyzed the epidemiological characteristics of all SARI cases reported over one year (2023–2024) in Rio Grande do Sul, Brazil. We also investigated the presence of respiratory viruses that were not included in the surveillance routine. For this purpose, we analyzed a subset of 4,000 specimens that remained without a confirmed diagnosis after initial screening. This expanded analysis employed a respiratory virus panel guided by the differential diagnosis recommendations for SARI outlined by the World Health Organization (WHO) [12], along with previous reports of SARI cases associated with endemic human coronaviruses in the region [13].

Our findings revealed the occurrence of SARI cases caused by various respiratory viruses not commonly included in routine diagnostic surveillance. Our results highlight the potential of viruses such as hRV, hMPV, and seasonal coronaviruses to cause severe infections and, in some cases, result in death, particularly among the elderly population. Alongside this, we observed a shift in the seasonality patterns of the studied viruses, necessitating adaptive public health strategies to address this evolving scenario.

Materials and methods

Population and clinical-epidemiological data

This study included all the 14,816 SARI cases reported between February 2023 and February 2024 in Rio Grande do Sul State, Brazil. SARI cases were identified according to the case definition established by Brazil's national surveillance guidelines for influenza and COVID-19. Specifically, SARI was defined as an individual with influenza-like illness presenting with at least two of the following acute respiratory symptoms: fever, chills, sore throat, headache, cough, coryza, anosmia, or ageusia (in children, nasal obstruction in the absence of another specific diagnosis was also considered), along with at least one of the following severity indicators: dyspnea or respiratory distress, persistent chest pain, oxygen saturation ≤ 94% on room air, or cyanosis of the lips or face, and requiring hospitalization.

Data related to each case were retrieved from the SIVEP-Gripe database, the Brazilian SARI case reporting

and investigation information system. The database includes all cases reported statewide by hospitals, health units, and municipal health departments. Most cases are investigated at the State Public Health Laboratory (LACEN-RS), where specimens suspected of viral respiratory infection are submitted to real-time RT-PCR molecular testing for IAV, IBV, SARS-CoV-2, and RSV using the Allplex™ SARS-CoV-2/FluA/FluB/RSV assay (Seegene Technologies Inc, Seoul, South Korea), and influenza subtyping using the RT-PCR protocol from CDC [14]. All the 14,816 SARI cases reported in the period were included in subsequent epidemiological analyses. A wide range of demographic, clinical, and epidemiological variables of each case were collected from SIVEP-Gripe.

Molecular investigation of the other respiratory viruses

We selected a subset of clinical specimens from SARI cases that had tested negative at LACEN-RS for IAV, IBV, SARS-CoV-2, and RSV. These specimens were then subjected to expanded molecular testing to detect additional respiratory viruses, including, hPIV, hMPV, hRV, hBoV, hAdV, EV, B19V, and four seasonal coronaviruses (229E, OC43, NL63, and HKU1).

A Total of 4,000 negative specimens were selected using a probabilistic random sampling approach implemented through a custom R script. This script stratified the selection by epidemiological week, ensuring that the number of randomly selected specimens per week was proportional to the total number of SARI cases reported during that same week. This method aimed to preserve the temporal distribution of cases and minimize potential selection bias. The sample size was defined based on operational limitations, including laboratory capacity, availability of reagents, and human resource constraints, while still allowing for a representative overview of the circulation of other respiratory viruses among SARI cases.

These 4,000 nasopharyngeal swab specimens or tracheal aspirates selected had been collected from hospitalized patients, sent to LACEN-RS for molecular diagnosis. The RNA/DNA was extracted using 200 µl of the selected specimens through automated magnetic bead assay on the Loccus Extracta 96 equipment with the FASTA– Viral DNA and RNA kit (Loccus, São Paulo, Brazil), following the manufacturer's protocol. To minimize potential PCR inhibition and ensure sample quality, all samples had been stored at − 80 °C to preserve RNA/DNA integrity. For this study, we performed new RNA extractions for all included samples, and samples with inconclusive or failed amplification were re-tested to confirm results and minimize the risk of false negatives.

The expanded respiratory panel testing was performed using different RT-qPCR assays. First, all selected specimens were tested using the Multiplex VR1/VR2 Kit (Bio-Manguinhos, Rio de Janeiro, Brazil) for detection in VR1 of IAV, IBV, SARS-CoV-2 and hMPV; in VR2 for hRV, RSV, and hAdV, along with human RNase-P as an internal control. The reactions were performed according to the manufacturer's instructions, including the following reaction cycling conducted using the 7500 Real-Time PCR System (Applied Biosystems): 50 °C for 15 min, 95 °C for 2 min, followed by 40 cycles of 95 °C for 15 s and 60 °C for 40 s. All the assays included at least one negative control and positive controls available in the kits.

For detection of hPIV, hBoV, EV, hB19V, and the four seasonal coronaviruses, singleplex real-time RT-PCRs were performed, with primers and probes specific for each virus [15–19] (Supplementary Table 1.). Each reaction was performed in a total volume of 20 µL, consisting of 12.5 µL of Master Mix, 0.5 µl of SuperScript™ III One-Step RT-PCR enzyme, 0.5 µL of each primer (specific for each virus) at a concentration of 10 nM, 0.5 µL of each probe (specific for each virus) at 5 nM, and 5 µL of extracted nucleic acid, brought to a final volume of 20 µL with nuclease-free water. RT-qPCR assays were performed were performed on ABI 7500 Real-Time PCR System (Applied Biosystems, Foster City, CA, USA) and CFX Opus Real-Time PCR System (Bio-Rad, Hercules, CA, USA), under the thermal cycling conditions: 50 °C for 30 min (only for RNA viruses), followed by 95 °C for 2 min, and then 45 cycles of 90 °C for 15 s and 55 °C for 30 s. Specimens were considered positive when the Ct value was ≤ 40.

Human RNase-P was also used as an internal control. Specimens without detectable RNase-P were deemed inadequate and excluded from the study, as this may indicate issues with sample collection, storage, or the presence of PCR inhibitors that could affect test reliability.

Statistical analysis

This study involved a descriptive analysis of frequencies, proportions, and association tests, including the chi-square and Fisher's exact test. To assess potential risk factors for unfavorable outcomes or disease severity associated with infections by different viruses, we employed multivariable logistic regression models adjusting for confounders such as age and comorbidities. This approach was also used to evaluate the association between codetection of two or more viruses and clinical complications, including ICU admission and death. We calculated odds ratios and confidence intervals, adopting a significance level of 0.05. Post hoc comparisons were conducted using Tukey's HSD and Bonferroni correction to control for multiple comparisons. All statistical analyses were performed using RStudio software.

Results

Demographics of SARI in Rio Grande do Sul, Brazil

During the 12-month study period, 14,816 SARI cases were notified in the State of Rio Grande do Sul, Brazil (Supplementary Fig. 1). The majority of cases occurred in children aged 0–9 years (43.51%), followed by adults aged 60 years and older (39.35%). According to data from SIVEP-Gripe, 7,204 (48.6%) specimens from SARI cases were positive for at least one respiratory virus. Of these, 3,396 (22.9%) specimens were positive for SARS-CoV-2, 2,329 (15.7%) were positive for RSV, 1,124 (7.6%) were positive for influenza virus, and 355 (2.4%) were associated with other respiratory viruses. Among influenza-positive cases, 802 (71.4%) were caused by IAV, while 322 (28.6%) were IBV. Of the IAV cases, 436 were subtyped as H1N1, 39 as H3N2, and 327 remained untyped or were not reported. The remaining 7,803 (52.7%) SARI cases reported had an undefined etiological agent (Fig. 1). Codetection was observed in 191 cases, including 80 cases that were positive for both SARS-CoV-2 and RSV, 35 cases of IAV and RSV codetection, 31 of IBV and RSV, 37 of SARS-CoV-2 and IAV, and eight cases of SARS-CoV-2 and IBV codetection.

Influenza virus, SARS-CoV-2, and RSV incidence varied over the study period. The proportion of SARS-CoV-2 cases peaked in April 2023 and then declined until mid-September, when a new peak was observed. Influenza and RSV cases peaked in May 2023, followed by a decline in case numbers. In contrast, the proportion of SARI cases without etiological identification increased during winter, peaking in July, and exhibited the highest number of cases throughout the entire study period (Fig. 2A). Among the 14,816 cases, 2,103 resulted in death, corresponding to a lethality rate of 15%. SARS-CoV-2 infection contributed to the highest number of deaths, with a lethality rate of 26.5%, followed by influenza (14.3%) and RSV (2.2%). Among the 7,803 undiagnosed infections, 1,071 deaths were registered, leading to a lethality rate of 14.6% (excluding cases with missing data for calculation). Overall, SARI cases were mostly reported in male individuals (51.14%), aged from 0 to 9 years old (43.50%) (Table 1), and were reported in most of the Rio Grande do Sul state territory (Fig. 2B).

The distribution of SARS-CoV-2 and influenza infections was relatively even across age groups, with a slight increase in cases among individuals ≥ 60 years old. In contrast, 82.8% of RSV infections occurred in individuals aged 0 to 1 year. When assessing the case fatality rate, results show that, for SARS-CoV-2 infections, the fatality rate increases significantly with age, reaching 34.6% of

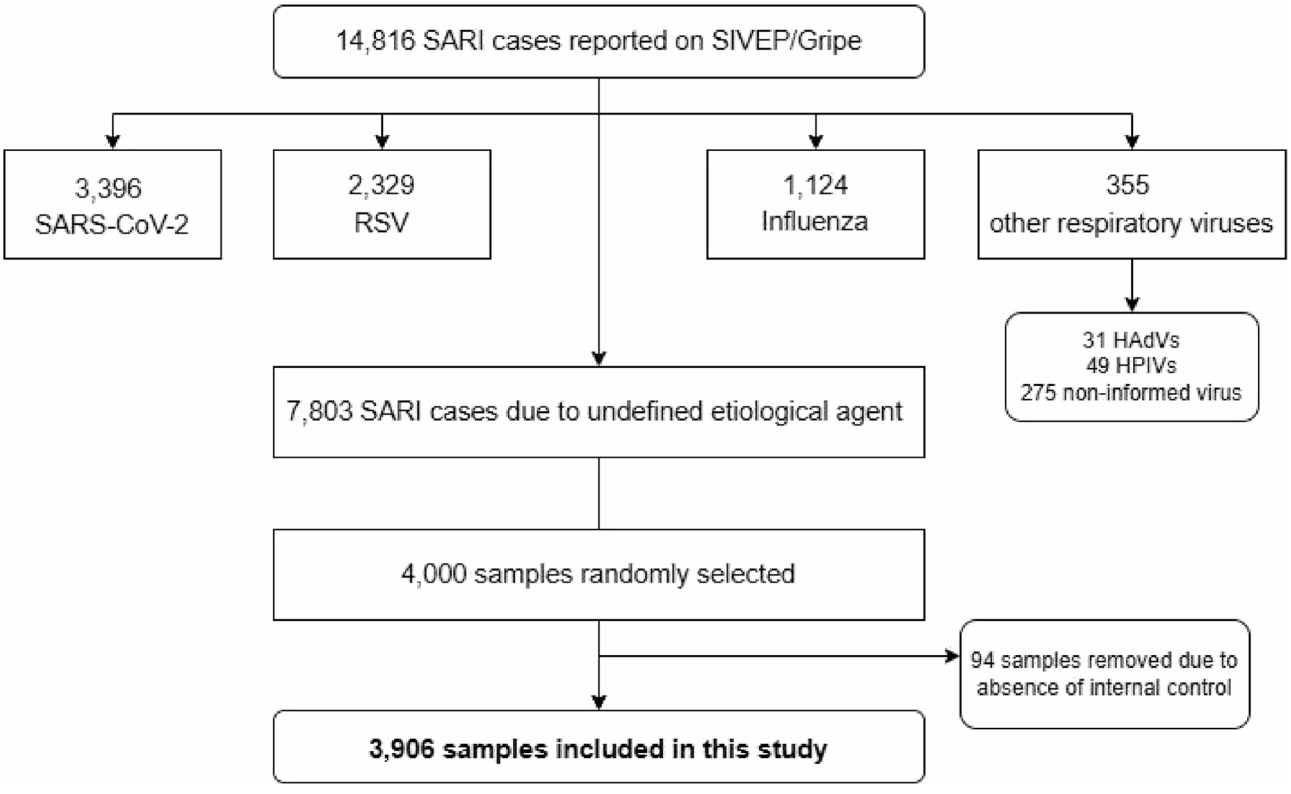


Fig. 1 Flowchart of the 14,816 SARI cases in Rio Grande do Sul, Brazil (February 2023– February 2024) reported in SIVEP-Gripe included in the study. The number of cases of SARS-CoV-2, influenza, and other respiratory viruses represents the positive cases detected at LACEN-RS or in other laboratories and reported in SIVEP-Gripe

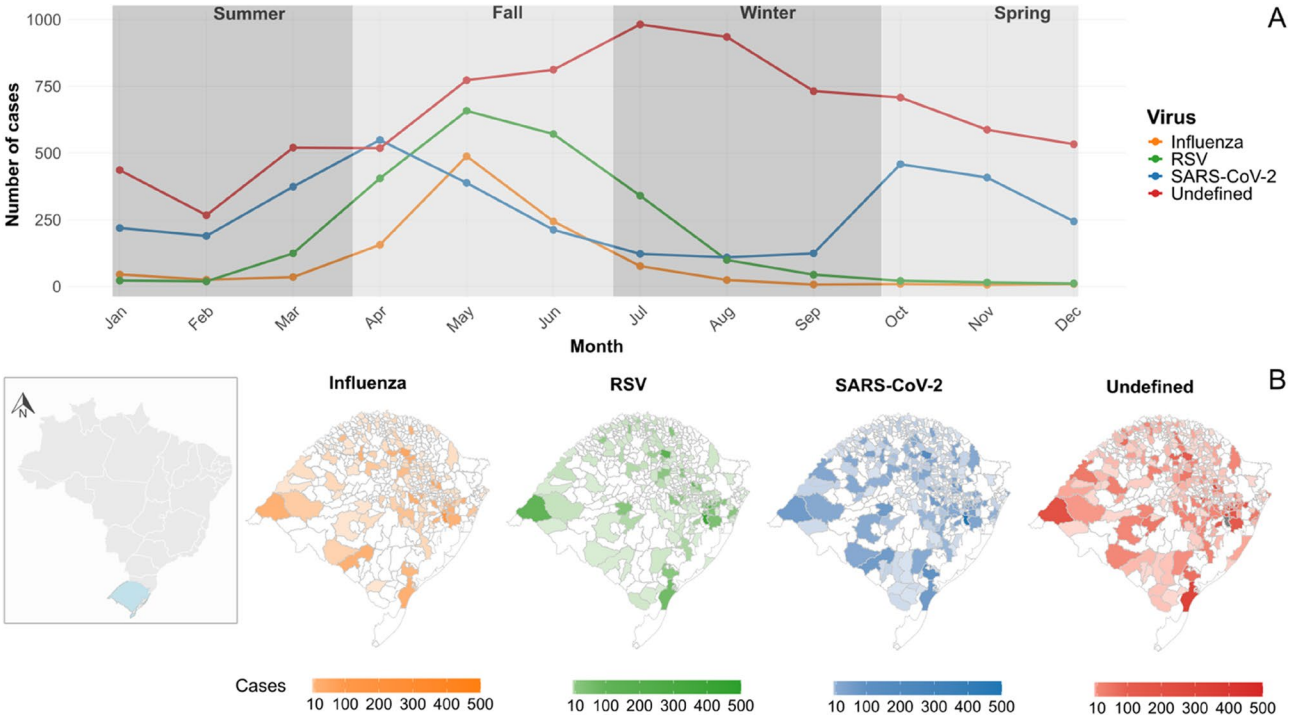


Fig. 2 Spatio-temporal distribution of SARI cases in Rio Grande do Sul, Brazil. **A** Seasonality patterns were observed for influenza, RSV, SARS-CoV-2, and for cases due to undefined etiological agents during the study period. **B** The map inside the square highlights the location of Rio Grande do Sul in Brazil. Each virus map represents the number of reported cases during the study period, along with the number of SARI cases attributed to undefined etiological agents in the same period

lethality for individuals ≥ 80 years old. A similar pattern is observed for influenza, though the age-related increase in lethality is less pronounced. The highest lethality rates for influenza infections were observed among individuals in the age groups 50–59 years (30.1%) and 60–69 years (34.26%). For RSV, the highest lethality rates were observed among individuals 0–9 years old. Among the SARI cases without an etiological diagnosis, we observed a significant increase in lethality rates starting at age 40, with the highest rate observed in individuals aged 80 years and older (34.4%) (Fig. 3).

We also calculated the percentage of patients diagnosed with each virus who required intensive care in intensive care units (ICU). Among patients with SARS-CoV-2 infections, 27.5% required ICU care, compared to 26.5% of those diagnosed with influenza, 26.9% with RSV, and 23.7% of individuals with SARI caused by other respiratory viruses. The analysis also showed that 25.4% of SARI patients with an undefined etiological agent were admitted to the ICU.

Expanded molecular detection on SARI cases due to an undefined etiological agent

Of the 4,000 specimens tested with the expanded Panel, 94 failed to amplify to the internal control hRNase-P and were excluded from the subsequent analysis. Among the 3,906 remaining specimens, at least one virus was detected in 1,741 specimens (44.6%). Additionally, 75 other specimens tested positive for influenza, SARS-CoV-2, or RSV, and were excluded from the subsequent analysis (these cases were initially reported in SIVEP-Gripe as negative and had been included among the 7,803 "undefined etiological agent" cases in Fig. 1). hRV was the most frequently detected virus, found in 1,102 (28,2% of the 3,906 cases), followed by hMPV in 308 (7.9%), hPIV in 277 (7.1%), hAdV in 237 (6.1%), and hBoV in 202 (5.2%) (Supplementary Table 2).

Additionally, 471 patients exhibited viral codetections, with two respiratory viruses detected in 368 (9.4% from 3,906 tested) cases, three in 81 (2.1%) cases, four in 18 cases (0.5%), and five viruses detected in four cases (0.1%). The most common cases of dual detections involved the following virus combinations: hRV and hAdV ($n = 71$), hRV and hBoV ($n = 49$), hRV and hMPV ($n = 41$), hRV and hPIV-3 ($n = 28$), and hBoV and hAdV ($n = 11$). The predominant cases of viral triple infections included hRV, hBoV, and hAdV ($n = 16$), hPIV-3, hAdV, and hRV ($n = 12$), hMPV, hRV, and hAdV ($n = 6$), and hMPV, hRV, and hBoV ($n = 6$). The complete list of simultaneous detection cases can be found in Supplementary Table 3. We investigated the potential association between co-infection and clinical severity or adverse outcomes. Logistic regression adjusting for age and comorbidities showed no significant association between

Table 1 Clinical and epidemiological characteristics of the 14,816 SARI cases analyzed in this study

Characteristics of SARI cases	All SARI cases	Influenza	SARS-CoV-2	RSV	SARI due to other respiratory viruses' infection	SARI due to an undefined etiological agent	Selected SARI due to an undefined etiological agent
Total	14816	1124	3396	2329	355	7803	3906
Sex (Chi-square statistic = 0.0444; *p*-value = 0.8331)							
Male	7577 (51.14%)	543 (48.31%)	1657 (48.79%)	1260 (54.10%)	191 (53.80%)	4020 (51.52%)	2004 (51.31%)
Female	7238 (48.85%)	581 (51.69%)	1739 (51.21%)	1069 (45.90%)	164 (46.20%)	3782 (48.47%)	1901 (48.67%)
Missing data	1 (0.01%)						1 (0.03%)
Age group (years) (Chi-square statistic = 1.691; p-value = 0.9891)							
0-9	6446 (43.51%)	318 (28.29%)	463 (13.63%)	2160 (92.74%)	265 (74.65%)	3373 (43.23%)	1661 (42.52%)
10-19	349 (2.36%)	61 (5.43%)	44 (1.30%)	13 (0.56%)	12 (3.38%)	224 (2.87%)	116 (2.97%)
20-29	275 (1.86%)	40 (3.56%)	82 (2.41%)	2 (0.09%)	4 (1.13%)	148 (1.90%)	68 (1.74%)
30-39	431 (2.91%)	71 (6.32%)	114 (3.36%)	4 (0.17%)	4 (1.13%)	240 (3.08%)	115 (2.94%)
40-49	585 (3.95%)	92 (8.19%)	154 (4.53%)	14 (0.60%)	3 (0.85%)	329 (4.22%)	163 (4.17%)
50-59	899 (6.07%)	96 (8.54%)	255 (7.51%)	30 (1.29%)	6 (1.69%)	517 (6.63%)	257 (6.58%)
60-69	1634 (11.03%)	147 (13.08%)	539 (15.87%)	32 (1.37%)	11 (3.10%)	921 (11.80%)	476 (12.19%)
70-79	1940 (13.09%)	157 (13.97%)	701 (20.64%)	34 (1.46%)	19 (5.35%)	1041 (13.34%)	526 (13.47%)
80 or more	2257 (15.23%)	142 (12.63%)	1044 (30.74%)	40 (1.72%)	31 (8.73%)	1010 (12.94%)	524 (13.42%)
Symptoms (Chi-square statistic = 4.273; p-value =0.9612)							
Fever	7441 (50.22%)	744 (66.19%)	1723 (50.74%)	1486 (63.80%)	239 (67.32%)	3881 (49.74%)	1947 (49.85%)
Cough	11868 (80.10%)	916 (81.49%)	2314 (68.14%)	2092 (89.82%)	284 (80.00%)	5942 (76.15%)	2936 (75.17%)
Sore Throat	1653 (11.16%)	174 (15.48%)	465 (13.69%)	68 (2.92%)	23 (6.48%)	658 (8.43%)	330 (8.45%)
Shortness of breath	15485 (104.52%)	774 (68.86%)	2278 (67.08%)	1785 (76.64%)	233 (65.63%)	6229 (79.83%)	3146 (80.54%)
Respiratory discomfort	11335 (76.51%)	633 (56.32%)	2046 (60.25%)	1572 (67.50%)	270 (76.06%)	4732 (60.64%)	2371 (60.70%)
Oxygen Saturation <95%	9993 (67.45%)	685 (60.94%)	2176 (64.08%)	1555 (66.77%)	272 (76.62%)	5442 (69.74%)	2768 (70.87%)
Diarrhea	1059 (7.15%)	82 (7.30%)	284 (8.36%)	124 (5.32%)	20 (5.63%)	535 (6.86%)	264 (6.76%)
Vomiting	1627 (10.98%)	126 (11.21%)	306 (9.01%)	219 (9.40%)	56 (15.77%)	724 (9.28%)	340 (8.70%)
Abdominal pain	774 (5.22%)	92 (8.19%)	234 (6.89%)	42 (1.80%)	11 (3.10%)	411 (5.27%)	205 (5.25%)
Fatigue	4933 (33.30%)	295 (26.25%)	1066 (31.39%)	335 (14.38%)	69 (19.44%)	1602 (20.53%)	805 (20.61%)
Loss of smell	177 (1.19%)	16 (1.42%)	60 (1.77%)	14 (0.60%)	0 (0.00%)	87 (1.11%)	32 (0.82%)
Loss of taste	217 (1.46%)	21 (1.87%)	86 (2.53%)	18 (0.77%)	12 (3.38%)	80 (1.03%)	38 (0.97%)
Outcome							
Cure	11933 (80.54%)	935 (83.19%)	2319 (68.29%)	2240 (96.18%)	336 (94.65%)	6275 (80.42%)	3041 (77.85%)
Death	2103 (14.19%)	156 (13.88%)	834 (24.56%)	50 (2.15%)	9 (2.54%)	1070 (13.71%)	588 (15.05%)
Missing data	780 (5.26%)	33 (2.94%)	243 (7.16%)	39 (1.67%)	10 (2.82%)	458 (5.87%)	277 (7.09%)
ICU							
Yes	3879 (26.18%)	298 (26.51%)	933 (27.47%)	627 (26.92%)	84 (23.66%)	1983 (25.41%)	1110 (28.42%)
No	10358 (69.91%)	793 (70.55%)	2305 (67.87%)	1653 (70.97%)	266 (74.93%)	5482 (70.26%)	2659 (68.07%)
Missing data	579 (3.91%)	33 (2.94%)	158 (4.65%)	49 (2.10%)	5 (1.41%)	338 (4.33%)	137 (3.51%)
Mechanical ventilatory support							
Invasive	1922 (12.97%)	178 (15.84%)	543 (15.99%)	275 (11.81%)	32 (9.01%)	1067 (13.67%)	626 (16.03%)
Non-invasive	7635 (51.53%)	599 (53.29%)	1754 (51.65%)	1725 (74.07%)	270 (76.06%)	4952 (63.46%)	2430 (62.21%)
Not required	2755 (18.59%)	306 (27.22%)	959 (28.24%)	256 (10.99%)	47 (13.24%)	1417 (18.16%)	706 (18.07%)
Missing data	2504 (16.90%)	41 (3.65%)	140 (4.12%)	73 (3.13%)	6 (1.69%)	367 (4.70%)	144 (3.69%)

co-infection and disease progression to death (b = 0.43, p = .115, OR = 1.54) or ICU admission (b = −0.16, p = .287, OR = 0.86), indicating that co-infection was not an independent predictor of worse clinical outcomes in this cohort.

Despite the 3,906 specimens included in the study being initially considered negative for influenza, SARS-CoV-2, and RSV, retesting using the VR1/VR2 Kit revealed 58 positive results for influenza, 43 for RSV, and 14 for SARS-CoV-2, resulting in a total of 115 cases (2.9%).

Seasonality of detected respiratory viruses

The respiratory viruses investigated in this study exhibited distinct seasonal patterns. hRV was detected throughout the year, with more pronounced activity

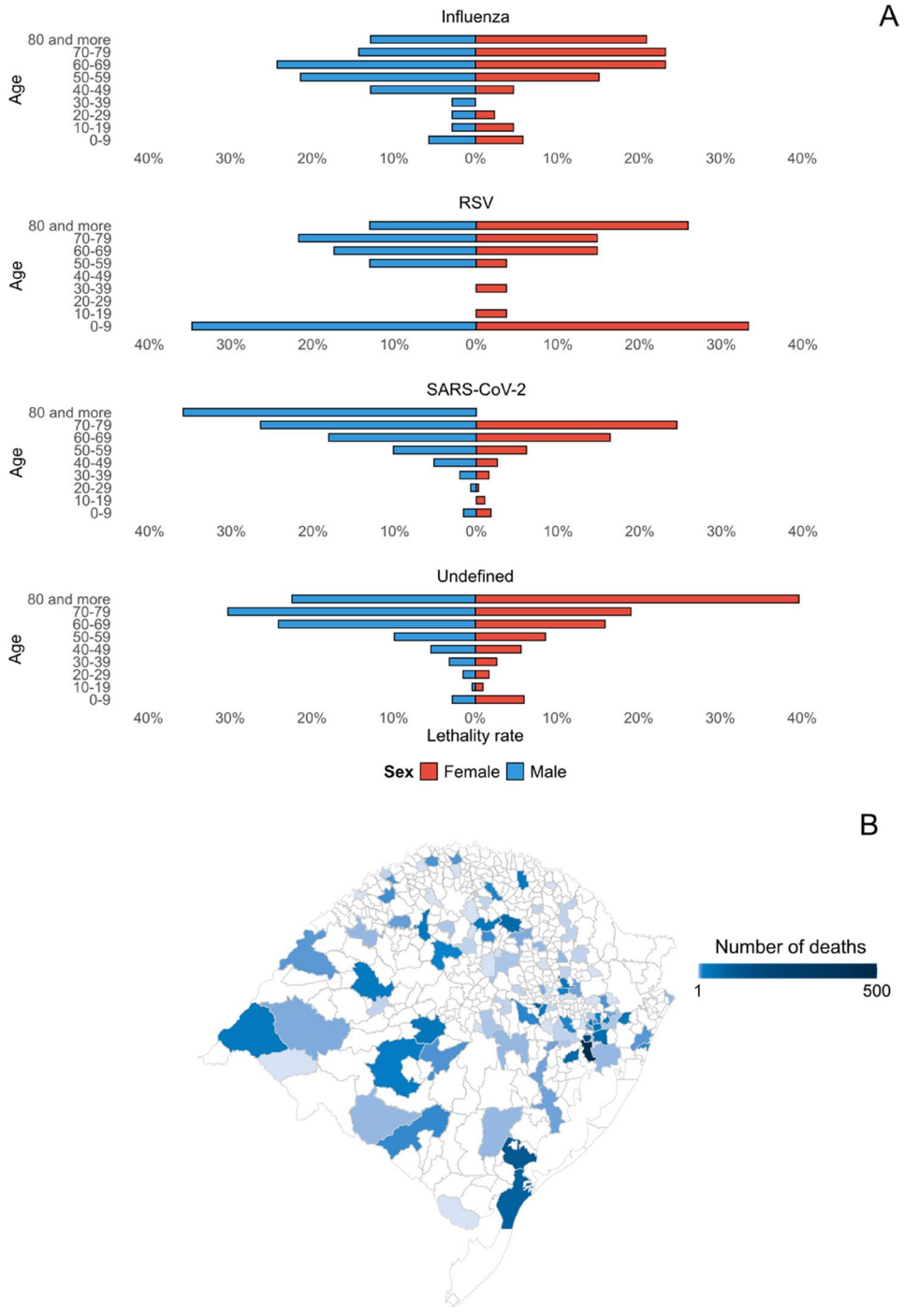


Fig. 3 (**A**) Lethality percentage for influenza, RSV, SARS-CoV-2, and cases with undefined etiological agents across different age groups and sex. The percentage was calculated by dividing the number of deaths by the total number of confirmed cases for each sex and age group, then multiplying by 100. **B** Geographic distribution of the 2,103 fatalities among the 14,816 analyzed SARI cases in Rio Grande do Sul, Brazil

peaks in the fall and winter. hMPV showed higher incidence at the end of fall and the beginning of winter. hAdV reached its peak of transmission during the winter months. hPIV was more frequently identified in winter and spring, and an increase in hBoV cases was observed at the end of fall, persisting through winter and into early spring. hCoV-NL63 presented a significant rise in cases between August 2023 and February 2024 (including winter, spring, and summer), tripling the number of cases compared to other months (Fig. 4). Notably, in January and February 2024, cases of most respiratory viruses, except hCoV-NL63 and hRV, showed a decline.

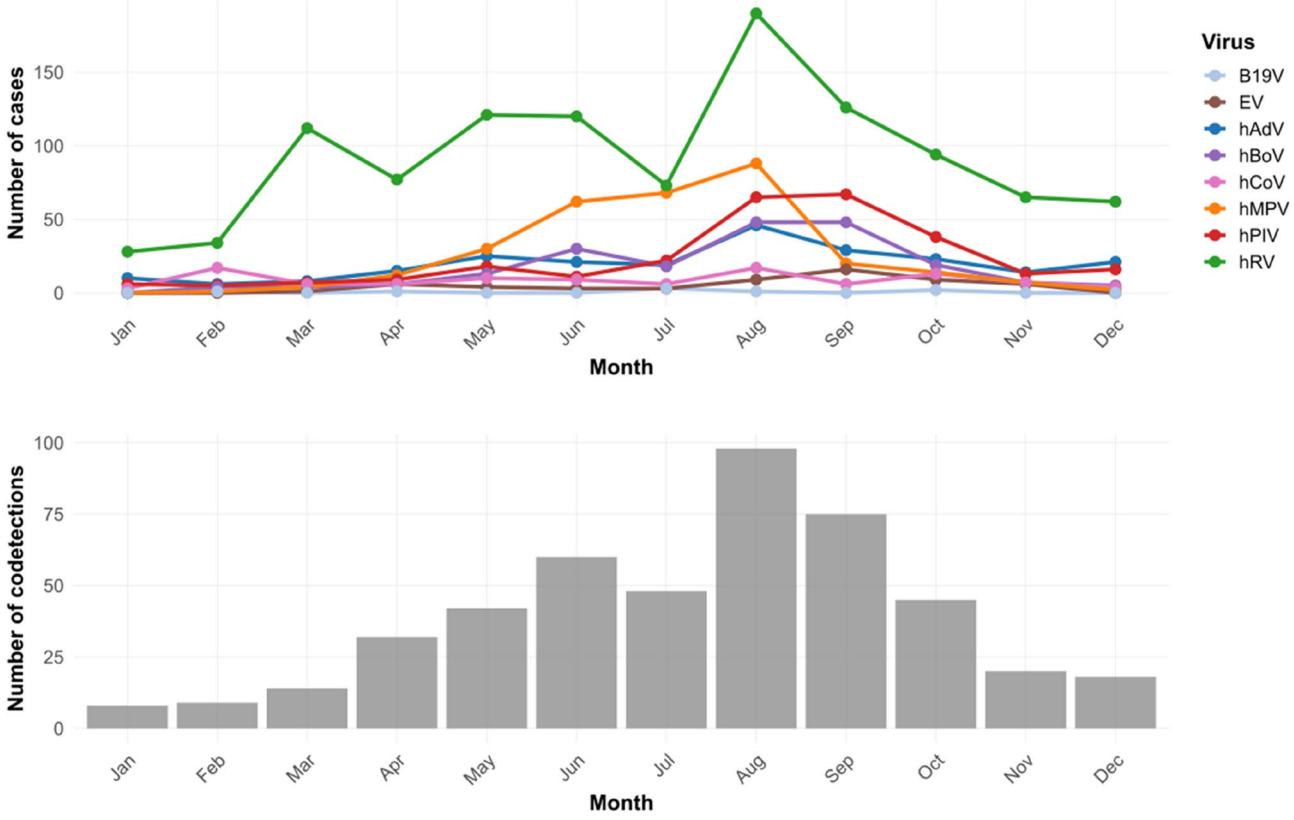


Fig. 4 Seasonality pattern observed among the respiratory viruses detected by the expanded RT-qPCR panel. hCoV includes the cases of the four endemic human coronaviruses investigated (229E, OC43, NL63, and HKU1). The gray bars represent the qPCR absolute number of codetections per month during the studied period

Characteristics of cases with detected respiratory viruses

Among the 1,741 positive cases, a higher prevalence of infection was observed in males (53.8%). Additionally, most infections caused by hRV, hMPV, hAdV, hBoV, hPIV, and EV occurred in children aged 0 to 9 years. Human coronaviruses NL63, HKU1, and 229E primarily affect children (0–9 years old) and individuals aged 70–79 years, while OC43 was detected in only one patient who was 98 years old. B19V infections were most prevalent among adults aged 50 to 69, while hRV was the most prevalent virus across all age groups (Supplementary Table 2). Among the cases that were positive only for hRV, one-third of the patients (287/758) had no comorbidities; 266 of these recovered, three died, and information was not available for the remaining 17 cases. In contrast, among the 472/758 patients with some underlying comorbidities, 401 recovered, 48 died; and for 25, there was no information available. A statistically significant association was observed between patient outcomes and the presence of underlying comorbidities (p-value < 0.00001). Similarly, most deaths attributed to the other viruses investigated occurred in patients with at least one underlying comorbidity. Accordingly, comorbidities were observed in all patients (100%) that died of either HAdV or B19V infection; in 91% of the deaths related To hPIV infection; in 90% of deaths associated with HBoV; in 80% of the deaths related to HCoV; in 73% of deaths related to hMPV infection; and in 50% of the deaths related to EV.

Among the 1,741 selected SARI cases with at least one virus detected, the primary symptoms observed were cough (83.5%), dyspnea (80.2%), oxygen saturation below 95% (68.5%), respiratory distress (59.2%), and fever in 55.6% of the cases. The need for ventilatory support was reported in 75.6% of cases, predominantly non-invasive type (66.6% of all cases). Among these patients, 23.4% required ICU hospitalization, and 7.6% died. Among patients who died, most of them (87.2%) had at least one underlying condition, with the most common being asthma, cardiovascular diseases, and other pneumonopathies (Supplementary Table 2).

Age-related susceptibility to death among viral infections

Among the 1,741 selected SARI cases with at least one virus detected, 133 (7.6%) individuals died. The distribution of deaths by age group (Fig. 5) showed that, in addition to ≥ 60 years old patients, the 0–9 years old age group recorded a significant number of deaths. hRV accounted

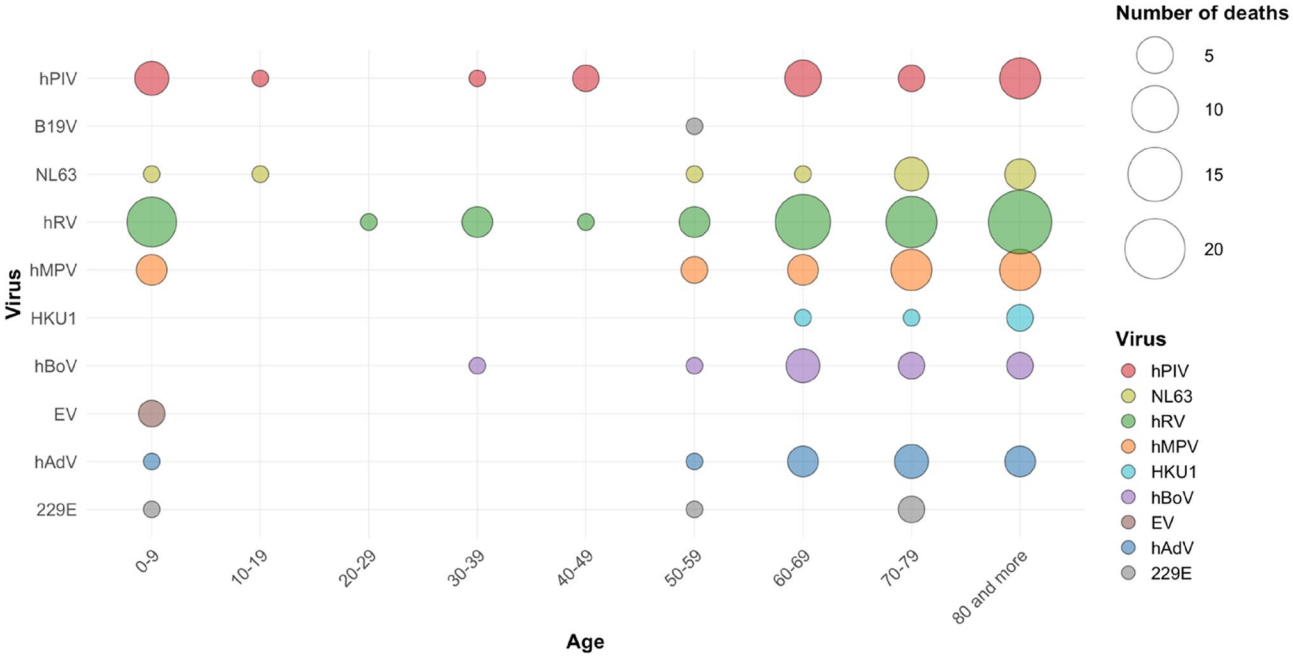


Fig. 5 Distribution of deaths in each age group across the different detected viruses among the 3,906 specimens using the expanded RT-qPCR Panel. The 229E, OC43, NL63, and HKU1 are the human coronaviruses (hCoV) investigated

for the majority of fatalities across all age groups, except for those aged 10 to 19 years. hMPV, hAdV, and hBoV were also associated with a notable number of deaths, particularly among children aged 0–9 years and adults aged 60–69, 70–79, and ≥ 80 years.

The analysis of the case fatality rate showed that the NL63 and HKU1 viruses exhibited the highest average lethality rates among the SARI cases investigated; however, these viruses were only detected in a few cases. Older age groups (60–69, 70–79, and ≥ 80 years old) had the highest average lethality rates. While hRV was the virus most associated with fatal cases, it demonstrated a moderate lethality rate across most age groups, with a higher lethality rate in adults aged 60–69 and ≥ 80 years old.

Specimens without detection of the etiological agent

Of the 3,906 specimens included in the study, the majority—53.6% (2,093/3,906)—tested negative for any respiratory virus. Among these 2,093 patients, 1,579 had comorbidities; 1,067 recovered, 114 remained hospitalized, 109 died from other causes, 16 had no recorded outcome, and 277 died. In contrast, among the 514 patients without comorbidities, 415 recovered, 37 were still hospitalized at the time of data collection, 12 died from other causes, six had no recorded outcome, and 44 died.

Discussion

In this study, we analyzed the etiological agents and clinical-epidemiological characteristics associated with SARI cases in Rio Grande do Sul, Brazil, between February 2023 and February 2024. The findings provide significant insights into the prevalence of several respiratory viruses not included in the surveillance routine in the region. A total of 14,816 SARI cases were reported in the state over a 12-month period. Notably, SARS-CoV-2 was the most detected virus, despite the absence of newly introduced lineages in Rio Grande do Sul state during the study period. RSV and influenza were the second and third most detected viruses, respectively. Among all cases of SARI, 7,803 were caused by undetermined etiological agents, representing approximately 52% of the total. These results underscore the complexity of respiratory infections and highlight the role of other respiratory viruses in severe infections leading to hospitalization.

To address this diagnostic gap, 4,000 randomly selected specimens from SARI cases without an identified etiological agent were subjected to in-depth molecular investigation. Results from this expanded testing revealed that 44.6% of these undiagnosed cases were associated with at least one respiratory virus not commonly included in the Brazilian surveillance routine. The viral detection in specimens initially considered negative for influenza, SARS-CoV-2, and RSV is an expected result due to the variation in sensitivity between the different molecular tests used. Such discrepancies are common in molecular studies, as different tests may have varying detection

sensitivity values, especially when detecting infections with low viral loads or co-infections [20].

The most frequently detected virus was hRV, followed by hMPV, hPIV, hAdV and hBoV. The frequencies observed in our study are consistent with those reported in previous studies conducted in West Africa and Vietnam before the COVID-19 pandemic [6, 21], and also align with global data on pediatric cases during the COVID-19 pandemic (2020–2023) [22, 23]. While hRV infections have historically been underestimated due to the perception that they only cause mild symptoms, often associated with the common cold [24], our results revealed the significant impact of hRV transmission, causing SARI in the general population, particularly among children and the elderly, as well as contributing to deaths. Accordingly, epidemiological studies show that hRV is frequently detected in adult and pediatric patients with respiratory tract infections and is becoming increasingly common in severe cases [25–27]. A similar scenario is observed for hMPV and hPIV, both of which have also been frequently underestimated despite their association with severe illness, as revealed by previous studies and by our results that showed several SARI cases caused by these viruses [28–30].

Additionally, recent outbreaks of hMPV, which pose significant risks to vulnerable populations such as children, the elderly, and immunocompromised individuals, have been reported across several countries, including China, Malaysia, India, the United Kingdom, and the United States. While the overall fatality rate of hMPV remains relatively low, this virus poses significant risks to vulnerable populations such as children, the elderly, and immunocompromised individuals, in whom it may cause severe complications such as pneumonia and acute respiratory failure [31]. Our findings highlight the substantial impact of hMPV infections in Southern Brazil, with over 300 reported SARI cases and more than 20 deaths in one year. This underscores the urgent need for strengthened surveillance in the state of Rio Grande do Sul, improved molecular diagnostics, and genomic research to inform public health strategies and mitigate the impact of hMPV.

hRV was the virus most frequently associated with fatal cases, and while the majority of cases were children [32, 33], the highest number of deaths was observed among individuals aged 80 and older. Additionally, patients infected solely by hRV who had underlying conditions were the most likely to die, highlighting the increased risk posed by comorbidities in the severity of hRV infections, a pattern also observed with the other viruses investigated. In addition to the substantial number of deaths associated with hRV among SARI cases, we observed a lethality rate of 7.4%. The rates were higher for hMPV (7.9%), hPIV (7.9%), and endemic coronaviruses (18.5%), while hBoV and hAdV had lower rates (5% and 5.1%, respectively). The observed rates emphasize the clinical severity of these pathogens and reflect their ability to cause severe disease, particularly in individuals with underlying comorbidities [34, 35].

Our results indicate that susceptibility to SARI cases varies significantly by age group. Children were found to be more susceptible to SARI caused by most of the viruses analyzed, while adults over 60 years of age faced a higher risk of death. This pattern aligns with observations reported in other studies and epidemiological scenarios, highlighting the age-specific vulnerabilities associated with respiratory infections [4, 36]. The high case fatality rates observed for human coronaviruses OC43 and HKU1 in our study may be influenced by the low number of SARI cases recorded for these viruses during the study period, particularly in younger age groups, as observed in previous studies in Rio Grande do Sul [13].

When examining seasonality, we observe that influenza and RSV peaked in the fall. At the same time, SARS-CoV-2 showed peaks in both fall and spring, coinciding with an increase in the notified SARI cases due to unknown etiological agents in the winter months. We also observed that in the seasonal distribution of SARI cases caused by hMPV, hRV, and hAdV, most cases peaked during the winter months. This is particularly important, as it reveals the significant contribution of these viruses to the total amount of respiratory infections and SARI cases in Rio Grande do Sul during winter. Every year, the winter season in Rio Grande do Sul is characterized by a substantial rise in respiratory infections and SARI cases, often leading to overcrowding of hospital capacity, mainly in the capital city, Porto Alegre.

The seasonality among the viruses studied here also draws attention due to the difference from what was expected regarding their distribution across the year. It is well known from several epidemiological studies in temperate regions that influenza, RSV, and coronaviruses have seasonal oscillation with incidence peaks occurring in the winter months (leading to them sometimes being called winter viruses) [37–39]. Conversely, viruses as hAdV, hBoV, hMPV, and hRV are commonly detected throughout the entire year (all-year viruses) [39–41], although hRV infection rates peak in spring and fall [42, 43].

Interestingly, our results revealed a different scenario for Rio Grande do Sul. Influenza and RSV peaked in the fall, while SARS-CoV-2 showed peaks in both fall and spring. Before the COVID-19 pandemic, influenza peaked in July 2018 and 2019. During the pandemic years (2020–2021), only a few cases were reported. However, influenza cases began To resurge in October 2022, with the incidence ultimately reaching its peak in the fall of 2023 [44, 45]. This pattern was also observed in the fall of 2024. On the other hand, we observed viruses, such as

hRV, hMPV, and hAdV reaching their peak in infections during the winter.

In recent years, a similar scenario of early circulation and an earlier peak in infections for influenza and RSV was observed in other regions of Brazil [46] and globally, following the COVID-19 pandemic [47]. Although multiple factors, such as viral interactions, changes in population behavior, and the effects of public health measures may contribute to this shift, they remain incompletely understood. Nevertheless, it is essential to adapt public health responses to mitigate the impact of these changing epidemiological trends [47]. Respiratory virus infections occurring during non-typical seasons can significantly impact public health services, increasing hospitalizations and higher demand for healthcare resources. Furthermore, prolonged or unexpected virus seasons may require more extensive interventions, such as expanded vaccination campaigns and strengthened public health measures. To effectively manage these shifting disease patterns, enhanced surveillance and adaptive strategies will be crucial in ensuring that public health systems can respond promptly and efficiently to emerging challenges.

The shared peaks of influenza and RSV during the same fall month also highlight an unusual epidemiological occurrence. Previous studies have demonstrated that influenza viruses and RSV usually do not share peak incidences, most probably due to competition for the host, so replication of a virus interferes with and limits replication of another virus [48]. Accordingly, during the 2009 influenza pandemic, hRV was believed to delay the onset of the European pandemic [49, 50]. Furthermore, a recent study has highlighted a strong negative interaction between influenza A and hRV at both the population and individual levels [51, 52]. Despite this evidence, codetections of respiratory viruses were found in several cases in our study, underscoring the complexity of respiratory infections.

Despite that respiratory viruses do not usually share circulation peaks, we observed several codetections, particularly between hRV and hAdV, hRV and hBoV, hRV and hMPV, and hRV and hPIV. This pattern of viral codetection is consistent with previous studies [53, 54], and, as demonstrated in a recent study, specimens testing positive for multiple agents by real-time PCR may contain infectious viruses [55]. This underscores the significant impact of multiple pathogens circulating simultaneously and their ability to infect the respiratory tract at the same time. Moreover, these several viral codetections stress the complex nature of respiratory infections, where multiple viruses can simultaneously or sequentially infect the respiratory tract, potentially leading to virus-virus interactions [56]. One virus's presence may influence another's infection dynamics, either enhancing or inhibiting its replication. These interactions emphasize the need for further studies, including in vitro experiments and cell culture models, to better understand the consequences of such viral interactions [57]. Such studies could clarify whether specific viruses alter host immune responses in ways that promote or suppress secondary infections, ultimately contributing to a more comprehensive understanding of respiratory virus dynamics.

Another critical consideration is that over half of the specimens analyzed in this study did not yield an identifiable etiological agent. Similar findings have been reported in other studies that tested SARI cases exclusively for viral pathogens, with approximately 60% of cases lacking a detectable viral etiology [6, 58]. This underdiagnosis issue highlights the complexity of SARI cases, which can be caused by a wide diversity of pathogens, including bacteria, fungi, or even non-infectious causes, such as allergic or inflammatory conditions. Many of these potential etiologies present with overlapping clinical symptoms, complicating differential diagnosis based solely on clinical criteria [59]. Moreover, our findings reflect the inherent limitations of PCR-based diagnostic panels. These include restricted pathogen coverage and variable sensitivity, which may be influenced by factors such as the timing of specimen collection, viral load, or sample quality [60].

To address this challenge and uncover the etiological agents responsible for these negative cases, next-generation sequencing (NGS)-based metagenomic detection emerges as a powerful and innovative tool for differential diagnostics [61]. While metagenomics may not be feasible for routine diagnostics in low- and middle-income countries like Brazil due to high costs, infrastructure requirements, and technical expertise, it can still serve as a powerful sentinel surveillance tool [62]. By strategically implementing metagenomics in selected sentinel sites or reference laboratories, it is possible to gain valuable insights into circulating pathogens, detect emerging threats, and guide public health responses without the need for widespread deployment.

Our study highlights the crucial role of epidemiological and molecular surveillance in monitoring the spread of respiratory viruses, elucidating their interactions and seasonal patterns, and contributing to the detection of potential outbreaks or increased healthcare demands. Additionally, our results stress the significant role of viruses other than SARS-CoV-2, influenza, and RSV in causing severe respiratory infections. These findings underscore the need for broader surveillance and diagnostic efforts to address the full spectrum of pathogens contributing to severe respiratory disease. Moreover, the observed shifts in viral seasonality can have important implications for public health services, leading to an increased demand for healthcare resources and challenging the existing capacity.

Despite the significance of our findings, this study has important limitations that should be acknowledged. First, it relies on data from the SIVEP-Gripe database, a national surveillance system for severe acute respiratory infections. As a secondary data source, SIVEP-Gripe is susceptible to data entry and completion errors, which may compromise the accuracy and completeness of the recorded information. Additionally, known limitations and potential biases associated with this database include reporting delays, underreporting, and possible inconsistencies in case definitions across different reporting hospitals, all of which may affect the reliability of the data. In this study, missing data were observed in key clinical outcome variables, including outcome status, which was absent in 5.2% of all reported SARI cases.

Additionally, samples received at the public health laboratory as suspected SARI cases are routinely tested only for respiratory viruses, and bacterial pathogens are not reported in the SIVEP-Gripe system, which was originally implemented to monitor influenza virus circulation in Brazil. This represents an important limitation, as certain bacterial genera can cause clinical symptoms that overlap with those of viral respiratory infections. Another limitation concerns the potential cross-reactivity between primers targeting hRV and EV due to their genetic similarity. Although we used an assay designed to detect EV species, this test did not allow for typing or differentiation of specific EV species, which may have led to misclassification or reduced specificity in distinguishing between these two virus groups. Additionally, while we detected a substantial proportion of hRV, RSV, hMPV in our samples, we were unable to classify these viruses into subtypes or genotypes due to the lack of sequence data and the scope of the current study.

The viral panel used in this study was based on protocols defined by the Brazilian Ministry of Health and did not include hPIV4, which has historically received limited attention due to its association with milder illness. While the exclusion of hPIV4 reflects a logistical limitation of the surveillance protocol, we recognize the importance of comprehensive viral monitoring that includes this virus. Furthermore, important clinical and epidemiological data, such as vaccination status, history of prior respiratory illness, and socioeconomic factors, were not consistently available and therefore were not included in our analysis. These variables could provide valuable insights into risk factors for severe disease and enhance etiological understanding. Future studies should aim to incorporate these factors to enable a more comprehensive assessment of SARI cases.

Finally, the observed scenario reveals the need for more extensive and intensive interventions, including expanded vaccination campaigns, enhanced surveillance systems, and strengthened public health measures. For instance, promoting the consistent use of masks among individuals who interact with elderly populations could be a valuable strategy. In the case of hRV, for example, the high proportion of infected children may lead to unintentional transmission to older adults, who are at greater risk of severe outcomes due to their increased vulnerability. Such targeted measures could help reduce the disproportionate impact on elderly populations and prevent avoidable fatalities. These steps are essential to mitigate the impact of respiratory viruses on vulnerable populations and ensure a more effective public health response.

Abbreviations

B19V	Human parvovirus B19
229E	Endemic human coronavirus 229E
COVID-19	Coronavirus disease 2019
Ct	Cycle threshold
EV	Respiratory enterovirus
hAdV	Human adenovirus
hBoV	Human bocavirus
hMPV	Human metapneumovirus
hPIV	Human parainfluenza viruses
hRV	Human rhinovirus
HKU1	Endemic human coronavirus HKU1
IAV	Influenza virus A
IBV	Influenza virus B
ICU	Intensive care unit
LACEN-RS	Rio Grande do Sul State Public Health Laboratory
µL	Microliter
nM	Nanomolar
NL63	Endemic human coronavirus NL63
OC43	Endemic human coronavirus OC43
PCR	Polymerase chain reaction
RSV	Respiratory syncytial virus
RT-PCR	Reverse transcription polymerase chain reaction
SARI	Severe acute respiratory infection
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
SIVEP-Gripe	Brazilian Influenza Epidemiological Surveillance Information System
WHO	World Health Organization

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12879-025-11458-5.

Supplementary Material 1. Supplementary Table 1. Primer and probe sequences for the detection of hPIV, hBoV, EV, hB19V, and the four seasonal coronaviruses, along with their respective sources. Supplementary Table 2. Clinical and epidemiological characteristics of the 3,906 SARI cases due to unknown etiological agents tested in this study. Supplementary Table 3. Simultaneous detection of two or more respiratory viruses investigated among 4,000 specimens from SARI cases.

Supplementary Material 2. Supplementary Figure 1. Spatio-temporal distribution of SARI cases in Rio Grande do Sul, Brazil. Each map shows the number of reported SARI cases during the study period, including cases attributed to undefined etiological agents. Additionally, the maps indicate the number of selected SARI cases tested using the expanded viral panel and the number of those cases that tested positive for at least one virus.

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Clinical trial

Not applicable.

Authors' contributions
APR: Conceptualization, investigation, data curation, formal analysis, writing– original draft, visualization, writing– review and editing. MCB: Investigation, data curation, formal analysis, writing– original draft, visualization, writing– review and editing. TSG: Investigation, data curation, formal analysis, writing– original draft, visualization, writing– review and editing. FMS: Investigation, data curation, formal analysis, writing– original draft, visualization, writing– review and editing. FG: Investigation, data curation, formal analysis, writing– original draft, visualization, writing– review and editing. LFB: Investigation, data curation, formal analysis, writing– original draft, visualization, writing– review and editing. TRMM: Investigation, data curation, formal analysis, writing– original draft, visualization, writing– review and editing. LGM: Formal analysis, methodology, writing– review and editing. RPM: Formal analysis, methodology, writing– review and editing. CP: Formal analysis, methodology, writing– review and editing. AC: Methodology, writing– review and editing. TS: Data curation, software, validation, visualization, writing– review and editing. PCR: Formal analysis, visualization, writing– review and editing. GLW: Funding acquisition, writing– review and editing. RSS: Project administration, funding acquisition, supervision, resources, writing– review and editing. ABGV: Conceptualization, supervision, resources, project administration, writing– review and editing.

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Data availability
All data supporting the findings of this study are available within the paper and its Supplementary Information.

Declarations

Ethics approval and consent to participate
The study was approved by the Ethical Committees of Escola de Saúde Pública do Rio Grande do Sul (CAAE 67181123.1.0000.5312) and Universidade Federal de Ciências da Saúde de Porto Alegre (CAAE 75118217.9.0000.5345), in accordance with the Declaration of Helsinki. This study was conducted using fully anonymized residual clinical samples. Written informed consent to participate was not required, as the research involved no identifiable human data and posed minimal risk. The informed consent was waived by the Escola de Saúde Pública do Rio Grande do Sul and Universidade Federal de Ciências da Saúde de Porto Alegre ethical committees in accordance with Brazilian national regulations (CNS Resolution Nº 510/2016).

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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References

1. Cui C, Timbrook TT, Polacek C, Heins Z, Rosenthal NA. Disease burden and high-risk populations for complications in patients with acute respiratory infections: a scoping review. Front Med. 2024;11: 1325236.
2. Lozano R, Naghavi M, Foreman K, Lim S, Shibuya K, Aboyans V, et al. Global and regional mortality from 235 causes of death for 20 age groups in 1990 and 2010: a systematic analysis for the global burden of disease study 2010. Lancet (London, England). 2012;380:2095.
3. He Y, Liu WJ, Jia N, Richardson S, Huang C. Viral respiratory infections in a rapidly changing climate: the need to prepare for the next pandemic. EBioMedicine. 2023;93:104593.
4. GBD 2016 Lower Respiratory Infections Collaborators. Estimates of the global, regional, and National morbidity, mortality, and aetiologies of lower respiratory infections in 195 countries, 1990–2016: a systematic analysis for the global burden of disease study 2016. Lancet Infect Dis. 2018;18:1191.
5. Klink T, Rankin DA, Piya B, Spieker AJ, Faouri S, Shehabi A, et al. Evaluating the diagnostic accuracy of the WHO severe acute respiratory infection (SARI) criteria in middle Eastern children under two years over three respiratory seasons. PLoS One. 2020;15:e0232188.
6. Alroy KA, Do TT, Tran PD, Dang TQ, Vu LN, Le NTH, et al. Expanding severe acute respiratory infection (SARI) surveillance beyond influenza: the process and data from 1 year of implementation in Vietnam. Influenza Other Respir Viruses. 2018. https://doi.org/10.1111/irv.12571.
7. Pronyk PM, de Alwis R, Rockett R, Basile K, Boucher YF, Pang V, et al. Advancing pathogen genomics in resource-limited settings. Cell Genom. 2023;3: 100443.
8. Tracking global progress on preparedness for respiratory pandemics. 2023 report. 2025. https://www.who.int/publications/i/item/9789240110434. Accessed 2 Jul 2025.
9. SRAG 2024–03/03. https://opendatasus.saude.gov.br/dataset/srag-2021-a-2024/resource/8cb52f73-0184-41d5-8a8f-87d8f415652c?inner_span=True. Accessed 12 Mar 2025.
10. SRAG 2024–03/03. https://opendatasus.saude.gov.br/dataset/srag-2021-a-2024/resource/8cb52f73-0184-41d5-8a8f-87d8f415652c?inner_span=True. Accessed 7 Mar 2025.
11. Painel de Hospitalizações de Síndrome Respiratória Aguda Grave - SRAG. https://ti.saude.rs.gov.br/srag/. Accessed 21 Feb 2025.
12. [No title]. https://iris.who.int/bitstream/handle/10665/331736/WHO-2019-nCoV-SARI_toolkit-2020.1-eng.pdf?utm_source=chatgpt.com. Accessed 3 Jul 2025.
13. da Veiga ABG, Martins LG, Riediger I, Mazetto A, do Carmo Debur M, Gregianini TS. More than just a common cold: endemic coronaviruses OC43, HKU1, NL63, and 229E associated with severe acute respiratory infection and fatality cases among healthy adults. J Med Virol. 2021;93:1002–7.
14. Jernigan DB, Lindstrom SL, Johnson JR, Miller JD, Hoelscher M, Humes R, et al. Detecting 2009 pandemic influenza A (H1N1) virus infection: availability of diagnostic testing led to rapid pandemic response. Clin Infect Dis. 2011;52: 1.
15. Zhou J, Peng Y, Peng X, Gao H, Sun Y, Xie L, et al. Human bocavirus and human metapneumovirus in hospitalized children with lower respiratory tract illness in Changsha, China. Influenza Other Respir Viruses. 2018;12(2):279.
16. Lu X, Chittaganpitch M, Olsen SJ, Mackay IM, Sloots TP, Fry AM, et al. Real-time PCR assays for detection of bocavirus in human specimens. J Clin Microbiol. 2006. https://doi.org/10.1128/JCM.00889-06.
17. Lu X, Holloway B, Dare RK, Kuypers J, Yagi S, Williams JV, et al. Real-Time reverse Transcription-PCR assay for comprehensive detection of human rhinoviruses. J Clin Microbiol. 2007;46:533.
18. Toppinen M, Norja P, Aaltonen LM, Wessberg S, Hedman L, Söderlund-Venermo M, et al. A new quantitative PCR for human parvovirus B19 genotypes. J Virol Methods. 2015. https://doi.org/10.1016/j.jviromet.2015.03.006.
19. Dare RK, Fry AM, Chittaganpitch M, Sawanpanyalert P, Olsen SJ, Erdman DD. Human coronavirus infections in rural Thailand: a comprehensive study using real-time reverse-transcription polymerase chain reaction assays. J Infect Dis. 2007;196:1321.
20. Fischer C, Mögling R, Melidou A, Kühne A, Oliveira-Filho EF, Wolff T, et al. Variable sensitivity of SARS-CoV-2 molecular detection in European expert laboratories: external quality assessment, June and July 2020. J Clin Microbiol. 2021. https://doi.org/10.1128/JCM.02676-20.
21. Lingani M, Cissé A, Ilboudo AK, Yaméogo I, Tarnagada Z. Patterns of non-influenza respiratory viruses among severe acute respiratory infection cases in Burkina Faso: a surveillance study. Influenza Other Respir Viruses. 2024;18: e13271.
22. Khales P, Razizadeh MH, Ghorbani S, Moattari A, Saadati H, Tavakoli A. Prevalence of respiratory viruses in children with respiratory tract infections during the COVID-19 pandemic era: a systematic review and meta-analysis. BMC Pulm Med. 2025;25:1–13.
23. Gill PJ, Kaziev CL, Mtaweh H, Kant T, Seaton C, Farrar DS, et al. Performance of the world health organization (WHO) severe acute respiratory infection (SARI) case definitions in hospitalized children and youth: cross-sectional study. Lancet Reg Health. 2025;44: 101034.
24. To KKW, Yip CCY, Yuen KY. Rhinovirus - from bench to bedside. J Formos Med Assoc. 2017. https://doi.org/10.1016/j.jfma.2017.04.009.

25. Pierangeli A, Scagnolari C, Selvaggi C, Verzaro S, Spina MT, Bresciani E, et al. Rhinovirus frequently detected in elderly adults attending an emergency department. J Med Virol. 2011;83:2043.
26. Jain S, Self WH, Wunderink RG, Fakhran S, Balk R, Bramley AM, et al. Community-acquired pneumonia requiring hospitalization among U.S. adults. N Engl J Med. 2015. https://doi.org/10.1056/NEJMoa1500245.
27. Role of rhinovirus load. In the upper respiratory tract and severity of symptoms in lung transplant recipients. J Clin Virol. 2015;64:1–5.
28. Kulkarni D, Cong B, Ranjini MJK, Balchandani G, Chen S, Liang J, et al. The global burden of human metapneumovirus-associated acute respiratory infections in older adults: a systematic review and meta-analysis. Lancet Healthy Longev. 2025. https://doi.org/10.1016/j.lanhl.2024.100679.
29. Gregianini TS, Seadi CF, Neto LDZ, Martins LG, Muller GC, Straliotto SM, et al. A 28-year study of human parainfluenza in Rio Grande do Sul, Southern Brazil. J Med Virol. 2019;91:1423.
30. Pan Y, Zhang Y, Shi W, Peng X, Cui S, Zhang D, et al. Human parainfluenza virus infection in severe acute respiratory infection cases in Beijing, 2014-2016: a molecular epidemiological study. Influenza Other Respir Viruses. 2017;11:564.
31. Kumar S, Subramaniam G, Senthil Kumar KJ. Human metapneumovirus (HMPV) outbreak: an urgent call for enhanced surveillance and public health action. J Med Virol. 2025;97:e70199.
32. da Costa Souza L, Bello EJM, dos Santos EM, Nagata T. Molecular and clinical characteristics related to rhinovirus infection in Brasília, Brazil. Braz J Microbiol. 2021;52:289.
33. Lu J, Wu T, Zeng Q, Chen Y, Liu Y, Wu D. Epidemiology of rhinovirus under the COVID-19 pandemic in Guangzhou, China, 2020. Immunity Inflamm Dis. 2022;10: e632.
34. Esper F, Boucher D, Weibel C, Martinello RA, Kahn JS. Human metapneumovirus infection in the United States: clinical manifestations associated with a newly emerging respiratory infection in children. Pediatrics. 2003;111:6 Pt 1.
35. Boon H, Meinders A-J, van Hannen EJ, Tersmette M, Schaftenaar E. Comparative analysis of mortality in patients admitted with an infection with influenza A/B virus, respiratory syncytial virus, rhinovirus, metapneumovirus or SARS-CoV-2. Influenza Other Respir Viruses. 2024;18:e13237.
36. Sentilhes AC, Choumlivong K, Celhay O, Sisouk T, Phonekeo D, Vongphrachanh P, et al. Respiratory virus infections in hospitalized children and adults in Lao PDR. Influenza Other Respir Viruses. 2013. https://doi.org/10.1111/irv.12135.
37. Tamerius J, Nelson MI, Zhou SZ, Viboud C, Miller MA, Alonso WJ. Global influenza seasonality: reconciling patterns across temperate and tropical regions. Environ Health Perspect. 2011;119:439–45.
38. Midgley CM, Haynes AK, Baumgardner JL, Chommanard C, Demas SW, Prill MM, et al. Determining the seasonality of respiratory syncytial virus in the United States: the impact of increased molecular testing. J Infect Dis. 2017;216:345–55.
39. Landes MB, Neil RB, McCool SS, Mason BP, Woron AM, Garman RL, et al. The frequency and seasonality of influenza and other respiratory viruses in Tennessee: two influenza seasons of surveillance data, 2010–2012. Influenza Other Respir Viruses. 2013;7:1122–7.
40. Bartlett N, Wark P, Knight D. Rhinovirus infections: rethinking the impact on human health and disease. Academic; 2019.
41. Bastien N, Brandt K, Dust K, Ward D, Li Y. Human bocavirus infection, Canada. Emerg Infect Dis. 2006;12:848–50.
42. Lee W-M, Lemanske RF Jr, Evans MD, Vang F, Pappas T, Gangnon R, et al. Human rhinovirus species and season of infection determine illness severity. Am J Respir Crit Care Med. 2012;186:886–91.
43. Monto AS. The seasonality of rhinovirus infections and its implications for clinical recognition. Clin Ther. 2002;24:1987–97.
44. Website. https://saude.rs.gov.br/upload/arquivos/201912/27150713-boletim-inluenza-2018-06-09-1.pdf
45. [No title]. https://saude.rs.gov.br/upload/arquivos/201911/27130006-boletim-semanal-flu-se-47.pdf. Accessed 21 Feb 2025.
46. Informe SE7. de 2025. https://www.gov.br/saude. Accessed 24 Mar 2025.
47. Global analysis of respiratory viral circulation. Timing of epidemics in the pre–COVID-19 and COVID-19 pandemic eras, based on data from the global influenza surveillance and response system (GISRS). Int J Infect Dis. 2024;144:107052.
48. Moriyama M, Hugentobler WJ, Iwasaki A. Seasonality of respiratory viral infections. Annu Rev Virol. 2020. https://doi.org/10.1146/annurev-virology-012420-022445.
49. Linde A, Rotzén-Ostlund M, Zweygberg-Wirgart B, Rubinova S, Brytting M. Does viral interference affect spread of influenza? Euro Surveillance: bulletin Europeen Sur les maladies transmissibles = European. Commun Dis Bull. 2009;14:19354.
50. Casalegno JS, Ottmann M, Duchamp MB, Escuret V, Billaud G, Frobert E, et al. Rhinoviruses delayed the circulation of the pandemic influenza A (H1N1) 2009 virus in France. Clin Microbiol Infect. 2010. https://doi.org/10.1111/j.1469-0691.2010.03167.x.
51. Nickbakhsh S, Mair C, Matthews L, Reeve R, Johnson PCD, Thorburn F, et al. Virus-virus interactions impact the population dynamics of influenza and the common cold. Proc Natl Acad Sci USA. 2019. https://doi.org/10.1073/pnas.1911083116.
52. Gonzalez AJ, Ijezie EC, Balemba OB, Miura TA. Attenuation of influenza A virus disease severity by viral coinfection in a mouse model. J Virol. 2018. https://doi.org/10.1128/JVI.00881-18.
53. Ito CRM, Santos MO, de Oliveira Cunha M, de Araújo KM, de Souza GRL, Rézio GS, et al. Rhinovirus infection and co-infection in children with severe acute respiratory infection during the COVID-19 pandemic period. Virulence. 2024;15:2310873.
54. Hanchi AL, Guennouni M, Ben Houmich T, Echchakery M, Draiss G, Rada N, et al. Changes in the epidemiology of respiratory pathogens in children during the COVID-19 pandemic. Pathogens. 2022;11:1542.
55. Kawase M, Suwa R, Sugimoto S, Kakizaki M, Kume Y, Chishiki M, et al. Evidence of the simultaneous replications of active viruses in specimens positive for multiple respiratory viruses. Microbiol Spectr. 2024. https://doi.org/10.1128/spectrum.01920-23.
56. Piret J, Boivin G. Viral interference between respiratory viruses. Emerg Infect Dis. 2022;28:273.
57. Deleveaux S, Clarke-Kregor A, Fonseca-Fuentes X, Mekhaiel E. Exploring the possible phenomenon of viral interference between the novel coronavirus and common respiratory viruses. J Patient-Cent Res Rev. 2023;10: 91.
58. Kelly ME, Gharpure R, Shivji S, Matonya M, Moshi S, Mwafulango A, et al. Etiologies of influenza-like illness and severe acute respiratory infections in Tanzania, 2017–2019. PLoS Glob Public Health. 2023;3:e0000906.
59. Al-Tawfiq JA, Zumla A, Gautret P, Gray GC, Hui DS, Al-Rabeeah AA, et al. Surveillance for emerging respiratory viruses. Lancet Infect Dis. 2014. https://doi.org/10.1016/S1473-3099(14)70840-0.
60. Qian Y-Y, Wang H-Y, Zhou Y, Zhang H-C, Zhu Y-M, Zhou X, et al. Improving pulmonary infection diagnosis with metagenomic next generation sequencing. Front Cell Infect Microbiol. 2021;10:567615.
61. Wang X, Stelzer-Braid S, Scotch M, Rawlinson WD. Detection of respiratory viruses directly from clinical samples using next-generation sequencing: a literature review of recent advances and potential for routine clinical use. Rev Med Virol. 2022;32: e2375.
62. Yek C, Pacheco AR, Vanaerschot M, Bohl JA, Fahsbender E, Aranda-Díaz A, et al. Metagenomic pathogen sequencing in resource-scarce settings: lessons learned and the road ahead. Front Epidemiol. 2022;2:926695.

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